

PRAYAS

2.0 2027

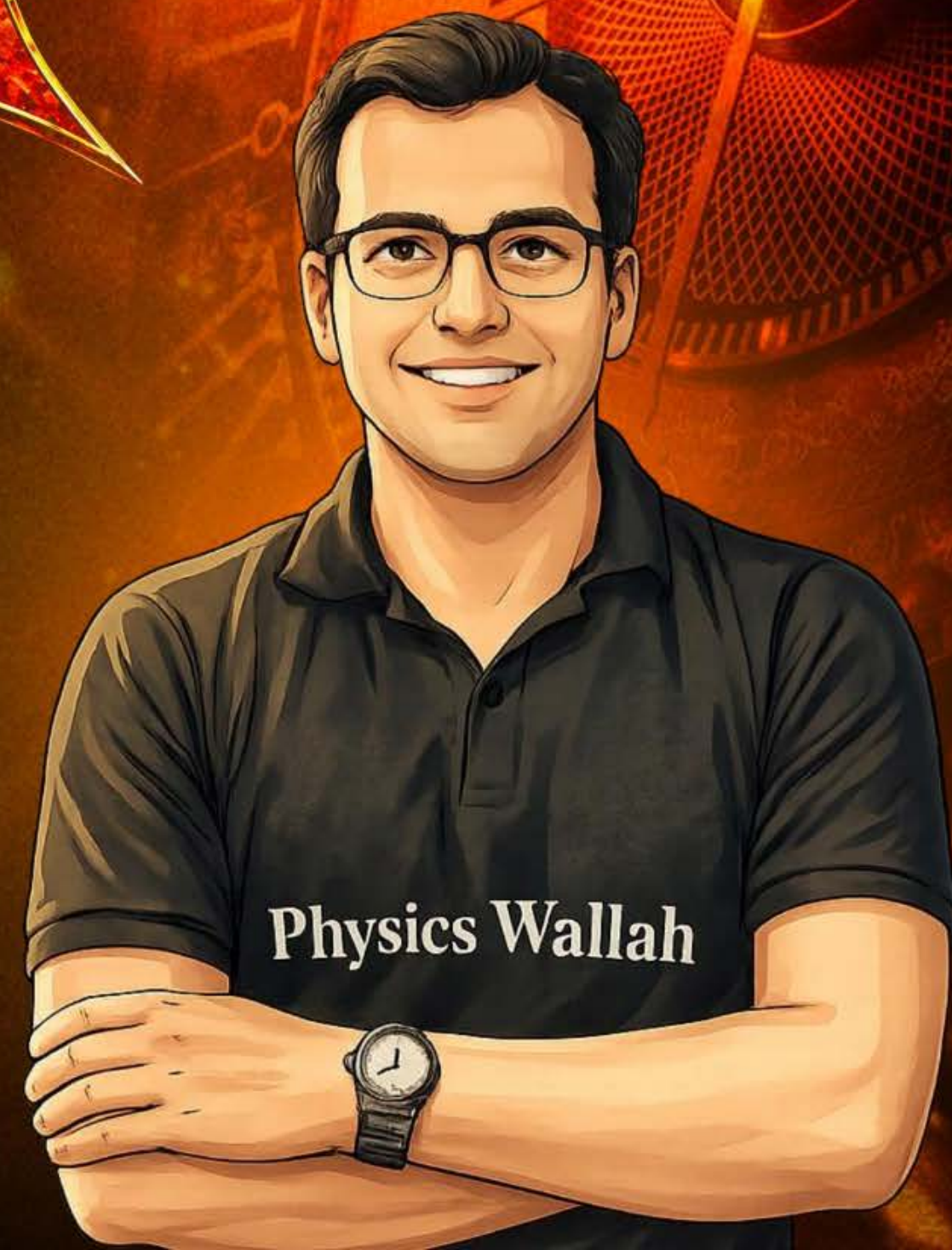
Mathematics

Lecture-10

Basic Mathematics

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Topics to be covered



1

Logarithm– Properties Contd...

2

Important Questions

Question

[Ans. (A, B, C)]



Question

The equation $x^{[(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5]} = 3\sqrt{3}$ has:

A Exactly 3 real solutions

B 2 Integral Solutions

C Exactly one irrational solution

D Non-Real Roots

[Ans. (A, B, C)]

$$(\log_3 x)^2 - \frac{9}{2} \log_3 x + 5 = \log_3 3\sqrt{3}$$

$$\left((\log_3 x)^2 - \frac{9}{2} \log_3 x + 5 \right) \log_3 x = \log_3 3^{3/2}$$

$$= 3/2 \times 1 = 3/2$$

Let $\log_3 x = t$

$$(t^2 - \frac{9}{2}t + 5)t = 3/2$$

$$(2t^2 - 9t + 10)t = 3$$

$$2t^3 - 9t^2 + 10t - 3 = 0$$

$t=1$ is a root by HIT

$$2t^2(t-1) - 7t(t-1) + 3(t-1) = 0$$

$$(t-1)(2t^2 - 7t + 3) = 0$$

$$3\sqrt{3} = 3^1 \cdot 3^{1/2} = 3^{3/2}$$

Brain Teaser



Solve the system of equations for x , y and z :

$$(\log_{2025} x) \log_{2025}(xyz) = 48$$

$$(\log_{2025} y) \log_{2025}(xyz) = 12$$

$$(\log_{2025} z) \log_{2025}(xyz) = 84$$

$$(\log x) \log(xyz) = 48$$

$$(\log y) \log(xyz) = 12$$

$$(\log z) \log(xyz) = 84$$

Add

$$\log(xyz) [\log x + \log y + \log z] = 144$$

$$\log(xyz) \log(xyz) = 144$$

$$[\log(xyz)]^2 = (12)^2$$

$$\log(xyz) = \pm 12$$

Case $\rightarrow 1$

$$\log(xyz) = 12$$

$$(\log x) \cdot 12 = 48$$

$$\log x = 4 \Rightarrow x = (2025)^4$$

$$\log y = 1 \Rightarrow y = (2025)^1$$

$$\log z = 7 \Rightarrow z = (2025)^7$$

Case $\rightarrow 2$

$$\log(xyz) = -12$$

$$x = (2025)^{-4}$$

$$y = (2025)^{-1}$$

$$z = (2025)^{-7}$$

$$\begin{aligned} \text{If } \sqrt{x^2} &= |x| \\ \text{If } x^2 &= y^2 \\ &\Rightarrow x = \pm y \end{aligned}$$

Example



Solve for x : $\log_{135} [(2^x - 7) + \log_{135}(2^x - 1)] = 1$

A -8 or 16

B 0 or 4

C 4 Only

D None of these

$$\log_{135} [(2^x - 7)(2^x - 1)] = 1$$

$$(2^x - 7)(2^x - 1) = (135)^1$$

$$2^x = t$$

$$(t - 7)(t - 1) = 135$$

$$t^2 - 8t + 7 = 135$$

$$t^2 - 8t - 128 = 0$$

$$t^2 - 16t + 8t - 128 = 0$$

$$(t - 16)(t + 8) = 0$$

$$t = 16 \text{ or } t = -8$$

$$2^x = 16 \text{ or } 2^x = -8$$

$$2^x = 2^4$$

$$\Rightarrow x = 4$$

$$\text{or } 2^x = (-2)^3$$

$$x = 3$$

$$2^x = -8$$

not possible

$2^x \rightarrow$ is never -ve

$$2^{-100} = \frac{1}{2^{100}} = +ve$$

$$2^0 = 1$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

Question



Solve for x : $x^{2 \log x} = 10x^2$

(where base of log is 10)

$$\log_{10} (x^{2 \log x}) = \log_{10} (10x^2)$$

$$2 \log_{10} x \log_{10} x = \log_{10} 10 + \log_{10} x^2$$

$$\text{Let } \log_{10} x = t$$

$$2t^2 = 1 + 2t$$

$$2t^2 - 2t - 1 = 0$$

$$t = \frac{2 \pm \sqrt{4 + 4 \times 2}}{2 \times 2}$$

$$t = \frac{2 \pm 2\sqrt{3}}{2 \times 2} = \frac{1 \pm \sqrt{3}}{2}$$

$$\log_{10} x = \frac{1 \pm \sqrt{3}}{2}$$

$$x = 10^{\frac{1+\sqrt{3}}{2}} \text{ or } 10^{\frac{1-\sqrt{3}}{2}} \checkmark$$

Question

[Ans. (511)]



Find the value of $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \dots \log_n(n+1) = 9$ find n .

$$\frac{\log 3}{\log 2} \cdot \frac{\log 4}{\log 3} \cdot \frac{\log 5}{\log 4} \dots \frac{\log(n+1)}{\log n}$$

$$\frac{\log(n+1)}{\log 2} = 9$$

$$\log_2(n+1) = 9$$

$$\Rightarrow n+1 = 2^9$$

$$n+1 = 512$$

$$n = 511$$

Question

Solve for x

$$1 + 2 \log_{(x+2)} 5 = \log_5 (x+2)$$

$$1 + 2 \cdot \frac{1}{t} = t$$

$$\frac{t+2}{t} = t$$

$$t+2 = t^2$$

$$t^2 - t - 2 = 0$$

$$(t-2)(t+1) = 0$$

$$t = 2 \text{ or } t = -1$$

$$\text{Let } \log_5 (x+2) = t$$

$$\Rightarrow \log (x+2)^5 = \frac{1}{t}$$

$$\log_5 (x+2) = 2 \text{ or } \log_5 (x+2) = -1$$

$$\Rightarrow x+2 = 25 \text{ or } x+2 = 5^{-1}$$

$$\boxed{x = 23}$$

Ans,,

$$x+2 = \frac{1}{5}$$

$$x = -2 + \frac{1}{5}$$

$$x = -\frac{9}{5}$$

Ans,,



RECAP

B.C.T →

P-7:

$$\log_b x = \frac{\log_c x}{\log_c b}$$

P-8:

$$\log_b a = \frac{1}{\log_a b}$$

Brain Teaser



Solve the equation $\log_x(9x^2) \cdot (\log_3 x)^2 = 4$

$$\log_x(3x)^2 \cdot (\log_3 x)^2 = 4$$

$$2 \log_x(3x) (\log_3 x)^2 = 4$$

$$2 (\log_x 3 + \log_x x) (\log_3 x)^2 = 4$$

$$2 (\log_x 3 + 1) (\log_3 x)^2 = 4$$

$$\text{Let } \log_3 x = t$$

$$\Rightarrow \log_x 3 = \frac{1}{t}$$

$$2 \left(\frac{1}{t} + 1 \right) t^2 = 4$$

$$t^2 \left(\frac{1}{t} + 1 \right) = 2$$

$$t + t^2 = 2$$

$$t^2 + t - 2 = 0$$

$$(t+2)(t-1) = 0$$

$$t = -2 \text{ or } t = 1$$

$$\log_3 x = -2 \text{ or } 1$$

$$\Rightarrow x = 3^{-2} \text{ or } 3^1$$

$$\Rightarrow x = \frac{1}{9} \text{ or } 3$$

Question



Find the Exact value of $\frac{\log_5 250}{\log_{50} 5} - \frac{\log_5 10}{\log_{1250} 5}$

$$(\log_5 250)(\log_5 50) - (\log_5 10) \log_5 (1250)$$

$$(3 + \log 2)(2 + \log 2) - (1 + \log 2)(4 + \log 2)$$

$$\log 2 = t$$

$$(3+t)(2+t) - (1+t)(4+t)$$

$$\cancel{t^2} + 6 + \cancel{5t} - (\cancel{t^2} + \cancel{5t} + 4)$$

$$6 - 4 = \boxed{2}$$



Advanced Properties of Logarithm



P-9:

Base power formula

$$\log_{a^k} m = \frac{1}{k} \log_a m$$

$$\log_{(b^k)} x = \frac{1}{k} \log_b x$$

Proof:

$$\frac{\log_b x}{\log_b (b)^k}$$

$$\frac{\log_b x}{k \log_b b} = \frac{1}{k} \log_b x = \text{RHS}$$

$$\log_{b^k} (x)^y = \frac{y}{k} \log_b x \quad \text{Ex} \rightarrow 2$$

$$\text{Ex} \rightarrow 1 \\ = \log_{512} 2$$

$$= \log_{2^9} 2 = \frac{1}{9} \log_2 2 \\ = \frac{1}{9}$$

$$\log_{27} 81 = \log_{3^3} (3)^4 \\ = \frac{4}{3} \log_3 3 \\ = \frac{4}{3}$$

Question

[Ans. (1)]



Find the value of $\log_2 10 - \log_8 125$.

$$\log_2 10 - \log_{(2^3)} (5)^3$$

$$\log_2 10 - \log_2 5$$

$$\log_2 \left(\frac{10}{5} \right)$$

$$= \log_2 2$$

$$= 1$$

Question

[Ans. (0)]



Find the value of $\frac{1}{\log_3 2} + \frac{2}{\log_9 4} - \frac{3}{\log_{27} 8} =$

$$\log_2 3 + 2 \log_4 9 - 3 \log_8 27$$

$$\log_2 3 + 2 \log_2 3 - 3 \log_2 3$$

$$3 \cancel{\log_2 3} - 3 \cancel{\log_2 3}$$

$$= 0$$

Logarithm of $32\sqrt[5]{4}$ to the base $2\sqrt{2}$ is

- A** 3.6 ✓
- B** 5
- C** 5.6
- D** None of these



Advanced Properties of Logarithm

P-10:

$$(a)^{\log_a N} = N$$

Proof:

Let $a^{\log_a N} = M$

To prove $M = N$

take $\log_a$ both sides

$$\log_a (a)^{\log_a N} = \log_a M$$

$$(\log_a N) \times 1 = \log_a M$$
$$\Rightarrow N = M$$

Ex $\rightarrow$

$$3^{\log_3 5} = 5$$

$$\pi^{\log_\pi 100} = 100$$

Ex $\rightarrow$

$$5^{2 \log_5 10} =$$

$$5^{\log_5 (10^2)}$$

$$= 10^2 = 100$$

Alt Proof $\rightarrow$

$$a^{\log_a N} = N^{\log_a a} = N^1 = N \checkmark$$

(P-11)

Question

[Ans. (...)]



$x^2 + 7^{\log_7 x} - 2 = 0$, then x is

A 2

B -2

C -1

D 1 ✓

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$\underbrace{x = -2}_{\text{Reject}} \text{ or } \underbrace{x = 1}_{\checkmark}$$

Question

Match the following:

Column-1		Column-2	
(1)	$\frac{1}{9}^{\log_3 7}$	(A)	$\frac{1}{27}$
(2)	$2^{-\log_{1/2} 7}$	(B)	$\frac{1}{2}$
(3)	$8^{-\frac{1}{\log_3 2}}$	(C)	$\frac{1}{49}$
(4)	$3^{-\log_3 2}$	(D)	7

$$= 2^{-1} = \left(\frac{1}{2}\right)$$

$$\begin{aligned}
 1) \quad \left(\frac{1}{9}\right)^{\log_3 7} &= \left(3^{-2}\right)^{\log_3 7} \\
 &= 3^{-2 \log_3 7} \\
 &= 3^{\log_3 (7)^{-2}} \\
 &= (7)^{-2} = \frac{1}{49} \\
 2) \quad 2^{-\log_{1/2} 7} &= 2^{-\log_{2^{-1}} 7} \\
 &= 2^{\log_2 7} = \textcircled{7} \\
 3) \quad 8^{-\log_2 3} &= \left(2^{-3 \log_2 3}\right) \\
 &= 2^{\log_2 (3)^{-3}} = (3)^{-3} = \frac{1}{27}
 \end{aligned}$$

Question

HW

[Ans. ()]



The value of $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$ is equal to

- A** 49
- B** 625
- C** 1090
- D** 890



Question

Solve:

(i) $2^{\log_2 x^2} - 3x - 4 = 0;$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$$x = 4 \text{ or } x = -1$$

Both accepted

(ii) $2^{2\log_2 x} - 3x - 4 = 0$

$$2^{\log_2 (x)^2} - 3x - 4 = 0$$

$$x^2 - 3x - 4 = 0 \rightarrow \begin{cases} x = 4 \checkmark \\ x = -1 \\ \text{Reject} \end{cases}$$

$$\log x^2 = 2 \log x$$

for common domain
which is $x \in (0, \infty)$

- ✓ A) Both correct
B) only $x = 4$
c) only $x = -1$

Question



[Ans. (A, B)]



$3^{\log_3^2 x} + x^{\log_3 x} = 162$, then x is

- A** 9
- B** $1/9$
- C** 10
- D** $1/10$

$$(x)^{\log_3 x} + (x)^{\log_3 x} = 162$$

$$2(x)^{\log_3 x} = 162$$

$$x^{\log_3 x} = 81$$

$$\log_3 (x)^{\log_3 x} = \log_3 81$$

$$(\log_3 x)(\log_3 x) = 4$$

$$(\log_3 x)^2 = 4$$

$$\log_3 x = \pm 2$$

$$x = 3^2 \text{ or } 3^{-2}$$

$$x = 9, 1/9$$

$$3^{t^2} = (3^t)^t$$

$$3^{\log_3^2 x} = (3^{\log_3 x})^{\log_3 x}$$

$$= (x)^{\log_3 x}$$



Advanced Properties of Logarithm

P-11:

$$a^{\log_b c} = c^{\log_b a}$$



$$8^{\log_{10} 5} = 5^{\log_{10} 8}$$

Question



$$\begin{aligned} & \overbrace{7^{\log_3 5} + 3^{\log_5 7} - 5^{\log_3 7} - 7^{\log_5 3}} \\ & \underbrace{5^{\log_3 7} + 7^{\log_5 3} - 5^{\log_3 7} - 7^{\log_5 3}}_{= 0} \end{aligned}$$

Question

Solve for x : $25^{\log_{10} x} = 5 + 4 \cdot x^{\log_{10} 5}$

$$5^{2x} = (5^x)^2$$

$5^t \rightarrow$ can't be -ve



$$5^{2 \log_{10} x} = 5 + 4 \underbrace{(x)^{\log_{10} 5}}_{(P-11)}$$

$$(5^{\log_{10} x})^2 = 5 + 4 (5)^{\log_{10} x^2}$$

$$\text{Let } 5^{\log_{10} x} = t$$

$$t^2 = 5 + 4t$$

$$t^2 - 4t - 5 = 0$$

$$(t-5)(t+1) = 0$$

$$t = 5 \text{ or } t = -1$$

$$5^{\log_{10} x} = 5 \text{ or } -1$$

(x)

$$5^{\log_{10} x} = 5^1$$

$$\log_{10} x = 1$$

$$x = 10^1 = 10 \checkmark$$



Advanced Properties of Logarithm

P-12:

$$a^{\sqrt{\log_a b}} = b^{\sqrt{\log_b a}}$$

*** Proof: LHS:
$$a^{\left(\frac{\sqrt{\log_a b} \cdot \sqrt{\log_a b}}{\sqrt{\log_a b}} \right)}$$

$$a^{\frac{\log_a b}{\sqrt{\log_a b}}}$$

p-10
$$a^{(\log_a b) \sqrt{\log_b a}} = (b)^{\sqrt{\log_b a}} = \text{RHS}$$

$$a^{\sqrt{\log_a b}} = b^{\sqrt{\log_b a}}$$

M-2 (Proof →)

Let $a^{\sqrt{\log_a b}} = x$

$\log_a$ both sides ✓

Ex:
$$7^{\sqrt{\log_7 2}} = 2^{\sqrt{\log_2 7}}$$

$$10^{\sqrt{\log_{10} 5}} = 5^{\sqrt{\log_5 10}}$$

If α, β are the roots of the equation

$$x^2 - x(5 + 3\sqrt{\log_3 5} - 5\sqrt{\log_5 3}) + 3(3^{\log_3 5} - 5^{\log_5 3} - 1) = 0$$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$ is :

A $3x^2 - 20x - 12 = 0$

B $3x^2 - 10x - 4 = 0$

C $3x^2 - 10x + 2 = 0$

D $3x^2 - 20x + 16 = 0$

$$x^2 - 5x - 3 = 0$$

$$\alpha + \beta = 5, \alpha\beta = -3$$

new sum :

$$\alpha + \frac{1}{\beta} + \beta + \frac{1}{\alpha}$$

$$\text{new product : } \left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$$

$$x^2 - (\text{sum})x + \text{product} = 0$$

$$\frac{(\log_3 5)^{1/3} (\log_3 5)^{2/3}}{3}$$

$$\left(3^{\log_3 5}\right) (\log_5 3)^{2/3}$$

$$(5) (\log_5 3)^{2/3}$$



Anti log

$$(\text{anti log}_b a) = (b)^a$$

$$\text{anti log}_b(x) = y$$

$$\Rightarrow x = \log_b y$$

$$y = b^x$$

$$\log_2 8 = 3$$

$$\Rightarrow 8 = (\text{anti log}_{\textcircled{2}} \textcircled{3})$$

$$\log_{10}(100) = 2$$

$$\Rightarrow \text{anti log}_{\textcircled{10}} \textcircled{2} = 100$$

$$\text{anti log}_3 5 = (3)^5$$

$$\text{anti log}_5 3 = (5)^3 = 125$$

Question

[Ans. (8)]



Let $x = (\text{anti log}_2 3) \cdot \log_3 2$, $y = \log_2(\log_3(\log_2 512))$ and $z = \log_5 3 \cdot \log_7 5 \cdot \log_2 7$, then xyz is equal to:

$$x = 8 \log_3 2$$

$$y = \log_2 \log_3 (9)$$

$$= \log_2 (2)$$

$$\boxed{y = 1}$$

$$z = \log_5 3 \log_7 5 \log_2 7$$

$$z = \frac{\log 3}{\log 2}$$

$$\begin{aligned} xyz &= 8 \log_3^2 \cdot 1 \times \log_2^3 \\ &= 8 \end{aligned}$$

Question



$\log_5(\sqrt[x]{5} + 125) = \log_5 6 + 1 + \frac{1}{2x}$, then x is

A -1

B $1/2$

C $1/4$

D No Solution

$$\log_5\left(\frac{\sqrt[x]{5} + 125}{6}\right) = 1 + \frac{1}{2x}$$

$$\frac{\sqrt[x]{5} + 125}{6} = 5^{1 + \frac{1}{2x}}$$

$$\frac{(5)^{1/x} + 125}{6} = 5 \cdot 5^{1/2x}$$

$$(5)^{1/x} + 125 = 30(5)^{1/2x} \rightarrow t$$

let $5^{1/2x} = t$

[Ans. (D)]



$$\Rightarrow 5^{1/x} = t^2$$

$$t^2 + 125 = 30t$$

$$t^2 - 30t + 125 = 0$$

$$(t - 25)(t - 5) = 0$$

$$t = 25 \text{ or } 5$$

$$5^{(1/2x)} = 5^2 \text{ or } 5^1$$

$$\frac{1}{2x} = 2 \text{ or } \frac{1}{2x} = 1$$

$$\Rightarrow x = \frac{1}{4} \text{ or } x = \frac{1}{2}$$

not in domain of $\sqrt[x]{5}$

$$f(x) = \sqrt[x]{5}$$

Domain: is $x = 2, 3, 4, 5, \dots$

$$f(x) = 5^{1/x}$$

$$x \in \mathbb{R} - \{0\}$$

Question

[Ans. (B,C)]



Domain: $x \in \mathbb{R} - \{0\}$

$$\log_5 \left(5^{\frac{1}{x}} + 125 \right) = \log_5 6 + 1 + \frac{1}{2x}, \text{ then } x \text{ is}$$

A -1

B ☒ 1/2

C ☒ 1/4

D No Solution

$$x = \frac{1}{2} \text{ or } x = \frac{1}{4}$$

$\Rightarrow$ Both are accepted.

$$x \in \mathbb{R} - \{0\}$$

If $\log_2 6 + \frac{1}{2x} = \log_2 (2^{\frac{1}{x}} + 8)$, then the value of x are

A $\frac{1}{4}, \frac{1}{3}$

B $\frac{1}{4}, \frac{1}{2}$

C $-\frac{1}{4}, \frac{1}{2}$

D $\frac{1}{3}, -\frac{1}{2}$

HW

If $(\log_{\beta}\alpha)^{2/3} + (\log_{\alpha}\beta)^{2/3} = 3$, then the value of $\left(\frac{(\log_{\beta}\alpha + \log_{\alpha}\beta)^2}{5}\right)$ is:

Let $\log_{\beta}\alpha = t$

$\log_{\alpha}\beta = \frac{1}{t}$

Given $t^{2/3} + \frac{1}{t^{2/3}} = 3$

Cube both sides

$$\left(t^{2/3} + \frac{1}{t^{2/3}}\right)^3 = 27$$

$$t^2 + \frac{1}{t^2} + 3\left(t^{2/3} + \frac{1}{t^{2/3}}\right) = 27$$

To Find $\frac{(t + \frac{1}{t})^2}{5}$

$$t^2 + \frac{1}{t^2} + 3 \times 3 = 27$$

$$t^2 + \frac{1}{t^2} = 27 - 9$$

$$t^2 + \frac{1}{t^2} = 18$$

$$\frac{t^2 + \frac{1}{t^2} + 2}{5} = ?$$

$$\text{Ans} = \frac{18 + 2}{5} = 4$$

$(x+1)^{\log_{10}(x+1)} = 100(x+1)$, then x is

- A** ✓ 99
- B** ✓ -0.9
- C** 0.9
- D** 10

$$\underbrace{\log_{10}(x+1)} \underbrace{\log_{10}(x+1)} = \log_{10} 100 + \log_{10}(x+1)$$

$$\log_{10}(x+1) = t$$

$$t^2 = 2 + t$$

$$t^2 - t - 2 = 0$$

$$(t-2)(t+1) = 0$$

$$t = 2 \text{ or } t = -1$$

$$\log_{10}(x+1) = 2 \text{ or } -1$$

$$x+1 = 10^2 \text{ or } x+1 = 10^{-1}$$

$$x+1 = 100 \text{ or } x+1 = \frac{1}{10}$$

$$x = 99 \text{ or } x = -\frac{9}{10}$$

Question

Simplify: $\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409} \cdot \left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - (125)^{\log_{25} 6} \right)$

$$\frac{(25 + 6\sqrt{6})(25 - 6\sqrt{6})}{409}$$

$$\frac{(25)^2 - (6\sqrt{6})^2}{409}$$

$$\frac{625 - 216}{409} = \frac{409}{409} = 1$$

[Ans. (1)]



$$\begin{aligned} 81^{\frac{1}{\log_5 9}} &= 81^{\log_9 5} \\ &= 9^{2 \log_9 5} \\ &= 9^{\log_9 5^2} \\ &= 5^2 = 25 \end{aligned}$$

$$\begin{aligned} 3^{\frac{3}{\log_3 \sqrt{6}}} &= (\sqrt{6})^3 = 6\sqrt{6} \end{aligned}$$

$$\begin{aligned} (\sqrt{7})^{\frac{2}{\log_7 25}} &= 7^{\log_7 25} \\ &= 25 \end{aligned}$$

$$\begin{aligned} (125)^{\log_{25} 6} &= 5^{3 \log_{5^2} 6} \\ &= 5^{\frac{3}{2} \log_5 6} \\ &= (6)^{3/2} = 6 \cdot 6^{1/2} \end{aligned}$$

Question

$$ab = \frac{\log 5}{\log 7} \cdot \frac{\log 3}{\log 5} = \log_7 3$$

[Ans. (D)]



If $\log_7 5 = a$, $\log_5 3 = b$ and $\log_3 2 = c$, then the logarithm of the number 70 to the base 225 is

- A** $\frac{1 - a + abc}{2a(1 + b)}$
- B** $\frac{1 - a - abc}{2a(1 + b)}$
- C** $\frac{1 + a - abc}{2a(1 + b)}$
- D** $\frac{1 + a + abc}{2a(1 + b)}$

$$\log_{225} (70) = \frac{\log_7 (70)}{\log_7 (225)} = \frac{\log_7 7 + \log_7 10}{2(\log_7 3 + \log_7 5)}$$

$$= \frac{1 + \log_7 5 + \log_7 2}{2(\log_7 3 + \log_7 5)}$$

$$\frac{1 + a + \log_7 2}{2(\log_7 3 + a)} \rightarrow abc$$

$$\frac{1 + a + \log_7 2}{2(\log_7 3 + a)}$$

$\downarrow$
ab

$$= \frac{1 + a + abc}{2(ab + a)}$$

$$= \frac{1 + a + abc}{2a(b + 1)}$$

$$abc = \frac{\log 5}{\log 7} \cdot \frac{\log 3}{\log 5} \cdot \frac{\log 2}{\log 3} = \log_7 2$$

Question

Solve for x :

$$2\log_x a + \log_{ax} a + 3\log_{a^2x} a = 0 \text{ is}$$

$$\frac{2}{\log_a x} + \frac{1}{\log_a(ax)} + \frac{3}{\log_a(a^2x)} = 0$$

$$\frac{2}{\log_a x} + \frac{1}{1 + \log_a x} + \frac{3}{2 + \log_a x} = 0$$

Let $\log_a x = t$

$$\frac{2}{t} + \frac{1}{1+t} + \frac{3}{2+t} = 0$$

$$\frac{2}{t} + \frac{1}{1+t} = -\frac{3}{2+t}$$

$$\frac{2(1+t) + t}{t(1+t)} = \frac{-3}{(2+t)}$$

$$\frac{2+3t}{t(1+t)} = \frac{-3}{(t+2)}$$

$$(2+3t)(t+2) = -3t(1+t)$$

$$2t+4+3t^2+6t = -3t-3t^2$$

$$6t^2 + 11t + 4 = 0$$

$$6t^2 + 8t + 3t + 4 = 0$$

$$(2t+1)(3t+4) = 0 \Rightarrow t = -\frac{1}{2}, -\frac{4}{3}$$

$x = a^{-1/2}$ or $a^{-4/3}$ Ans

The values of $6 + \log_{\frac{3}{2}} \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \dots \infty}}} \right)$ is:

$$y = \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \dots \infty}}}$$

$\rightarrow y$

$$y = \frac{1}{3\sqrt{2}} \sqrt{4 - y}$$

$$y^2 = \frac{1}{18} (4 - y)$$

$$18y^2 = 4 - y$$



Home Work Brain Teasers



Simplify : $5^{\log_{1/5}\left(\frac{1}{2}\right)} + \log_{\sqrt{2}} \frac{4}{\sqrt{7}+\sqrt{3}} + \log_{1/2} \frac{1}{10+2\sqrt{21}}.$

If $x = \log_a bc$, $y = \log_b ca$, $z = \log_c ab$, then $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} =$

A -1

B 1

C 0

D 3

If $a = \log_{12} 18$ and $b = \log_{24} 54$, then find the value of $ab + 5(a - b)$.

- A** -1
- B** 1
- C** 0
- D** 3

If a, b, c are distinct positive numbers, each different from 1, such that $[\log_b a \log_c a - \log_a a] + [\log_a b \log_c b - \log_b b] + [\log_a c \log_b c - \log_c c] = 0$, then $abc =$

- A** 1
- B** 2
- C** 3
- D** None of these

If $a = \log_2 3$ and $b = \log_3 5$ and $c = \log_7 2$, then find the value of $\log_{140} 63$ in terms of a, b, c .

If $a = \left(\sqrt{4 + \sqrt{15}} + \sqrt{4 - \sqrt{15}} - 2\sqrt{3 - \sqrt{5}} \right)$, $b = \left(\sqrt{17 + 4\sqrt{13}} - \sqrt{17 - 4\sqrt{13}} \right)$,
then the value of $\log_a b$ is:

Find the product of roots of the equation

$$\sqrt{2023}(x^{\log_{2023} x}) = x^2$$

Find x satisfying the equation

$$\log^2 \left(1 + \frac{4}{x} \right) + \log^2 \left(1 - \frac{4}{x+4} \right) = 2\log^2 \left(\frac{2}{x-1} - 1 \right).$$

Ans. $x = \sqrt{2}$ or $\sqrt{6}$

Find ' x ' satisfying the equation

$$4^{\log_{10} x + 1} - 6^{\log_{10} x} - 2 \cdot 3^{\log_{10} x^2 + 2} = 0.$$

Find the real solutions to the system of equations

$$\log_{10}(2000xy) - \log_{10} x \cdot \log_{10} y = 4$$

$$\log_{10}(2yz) - \log_{10} y \cdot \log_{10} z = 1$$

and $\log_{10}(zx) - \log_{10} z \cdot \log_{10} x = 0$

[Ans. $x = 1, y = 5,$
 $z = 1$ or $x = 100,$
 $y = 20, z = 100$]



If (x_1, y_1) and (x_2, y_2) are the solution of the system of equation $\log_{225} (x) + \log_{64} (y) = 4$
 $\log_x (225) - \log_y (64) = 1$, then find the value of $\log_{30}(x_1 y_1 x_2 y_2)$

Let $a = 3\sqrt{2}$ and $b = \frac{1}{5^{1/6}\sqrt{6}}$. If $x, y \in \mathbb{R}$ are such that

$$3x + 2y = \log_a (18)^{\frac{5}{4}} \text{ and}$$

$$2x - y = \log_b (\sqrt{1080}),$$

then $4x + 5y$ is equal to

Question [JEE Adv. 2018]

[Ans. (8)]



HW

The value of $((\log_2 9)^2)^{\frac{1}{\log_2(\log_2 9)}} \times (\sqrt{7})^{\frac{1}{\log_4 7}}$ is

Question [JEE Adv. 2013]

[Ans. A,B,C]



If $3^x = 4^{x-1}$, then $x =$

HW

A $\frac{2\log_3 2}{2\log_3 2 - 1}$

B $\frac{2}{2 - \log_2 3}$

C $\frac{1}{1 - \log_4 3}$

D $\frac{2\log_2 3}{2\log_2 3 - 1}$



Practice Problems on Log



Question

Find the Exact value of $\frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$

[Ans. (3)]



Question

Find the Exact value of $\frac{\log_5 250}{\log_{50} 5} - \frac{\log_5 10}{\log_{1250} 5}$

[Ans. (2)]



Question

Prove that $\frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2} = 3$



Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$. Find y

Question

Column-I

- (A) $\log_2 x = -\log_{\frac{1}{2}} 7$, then the value of x is
- (B) $\log_8 y = \frac{-1}{\log_3 2}$, then the value of y is
- (C) $\log_{\frac{1}{4}} z = \log_2 6$, then the value of z is
- (D) $\log_{\frac{1}{9}} w = \log_3 7$, then the value of w is

[Ans. (A) $\rightarrow$ S; (B) $\rightarrow$ R; (C) $\rightarrow$ Q; (D) $\rightarrow$ P]

Column-II

- (P) $\frac{1}{49}$
- (Q) $\frac{1}{36}$
- (R) $\frac{1}{27}$
- (S) 7
- (T) $\frac{1}{6}$



Question

[Ans. C]



Product of all the solution of equation $x^{\log_{10} x} = \left(100 + 2\sqrt{\log_2 3} - 3\sqrt{\log_3 2}\right) x$ is:

(A) $\frac{1}{10}$

(B) 1

(C) 10

(D) 100

Question

[Ans. B]

If $\log_7 2 = m$, then the value of $\log_{49} 28$ is:

(A) $2(1 + 2m)$

(B) $\frac{1 + 2m}{2}$

(C) $\frac{2}{1 + 2m}$

(D) $1 + m$

Question

$\frac{1}{\log_{\sqrt{bc}} abc} + \frac{1}{\log_{\sqrt{ca}} abc} + \frac{1}{\log_{\sqrt{ab}} abc}$ has the value equal to:

[Ans. B]

(A) $1/2$

(B) 1

(C) 2

(D) 4

Question

If $5x^{\log_2 3} + 3^{\log_2 x} = 162$, then logarithm of x to the base 4 has the value equal to:

[Ans. D]

(A) 2

(B) 1

(C) -1

(D) $3/2$

Question

If $\log(x + y) = \log 2 + \frac{1}{2}\log x + \frac{1}{2}\log y$, then:

[Ans. D]

(A) $x + y = 0$

(B) $xy = 1$

(C) $x^2 + xy + y^2 = 0$

(D) $x - y = 0$

Question

If $\log_2(\log_3(\log_4 x)) = 0$, $\log_4(\log_3(\log_2 y)) = 0$ and $\log_3(\log_4(\log_2 z)) = 0$, then the correct option is:

[Ans. B]

(A) $x > y > z$

(B) $x > z > y$

(C) $z > x > y$

(D) $z > y > z$

Question

The value of $\log_2 \left(\frac{1}{7^{\log_7 0.125}} \right)$, is:

[Ans. C]

- (A) 1 (B) 2 (C) 3 (D) 4

Question

Let x satisfies the equation $\log_3(\log_9 x) = \log_9(\log_3 x)$, then the product of the digits in x is:

[Ans. D]

- (A) 9 (B) 18 (C) 36 (D) 8

Question

If $\log_3(\log_2 a) + \log_{\frac{1}{3}}\left(\log_{\frac{1}{2}} b\right) = 1$, then the value of ab^3 is:

[Ans. C]

- (A) 9 (B) 3 (C) 1 (D) 1/3

Question

The value of $\log_{\frac{4}{3}} \left(\frac{56 + \sqrt{56 + \sqrt{56 + \sqrt{56 + \dots \infty}}}}{\sqrt{64 \sqrt{64 \sqrt{64 \dots \infty}}}} \right)$ is equal to:

[Ans. A]

- (A) 0 (B) 2 (C) 3 (D) 4

Question

The value of $\log_{\frac{1}{6}} 2 \cdot \log_5 36 \cdot \log_{17} 125 \cdot \log_{\frac{1}{\sqrt{2}}} 17$, is equal to:

[Ans. D]

- (A) -3 (B) -6 (C) 6 (D) 12

Question



(i) If $\log_4 10 = x$, $\log_2 20 = y$ and $\log_5 8 = z$.

Prove that $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$.

(ii) If $x = \log_a(bc)$, $y = \log_b(ca)$, $z = \log_c(ab)$.

Prove that $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$.

Question

(i) Prove that: $\frac{1}{1+\log_b a + \log_b c} + \frac{1}{1+\log_c a + \log_c b} + \frac{1}{1+\log_a b + \log_a c} = 1$.

(ii) Show that: $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{43} n} = \frac{1}{\log_{43!} n}$.

BT**[Ans. 469]**

If a, b, c are positive real numbers such that $a^{\log_3 7} = 27$; $b^{\log_7 11} = 49$ and $c^{\log_{11} 25} = \sqrt{11}$.

Find the value of $(a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2})$.

BT**[Ans. 5625]**

Positive numbers x, y and z satisfy $xyz = 10^{81}$ and $(\log_{10} x)(\log_{10} yz) + (\log_{10} y)(\log_{10} z) = 468$.

Find the value of $(\log_{10} x)^2 + (\log_{10} y)^2 + (\log_{10} z)^2$.



Let (x_1, y_1, z_1) and (x_2, y_2, z_2) are 2 set of solution satisfying the following system of equations $\log_{10}(2xy) = 4 + (\log_{10} x - 1)(\log_{10} y - 2)$
 $\log_{10}(2yz) = 4 + (\log_{10} y - 2)(\log_{10} z - 1)$, $\log_{10}(zx) = 2 + (\log_{10} z - 1)(\log_{10} x - 1)$
 such that $x_1 > x_2$, then match List-I with List-II and select the correct answer using the code given below the list.

List-I

(P) $\frac{y_1}{x_1}$

(Q) $\frac{z_1}{x_2}$

(R) $\frac{z_1 x_2}{z_2}$

(S) $\frac{y_2 + z_1}{x_2}$

List-II

(1) 2

(2) 100

(3) 1000

(4) 150

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$



Homework



❖ **DPP- 08**

Thank

You

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